MAP-Graph: Supplementary Material

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Organization and Relation to the Main Paper

This document supplies implementation and evaluation details omitted from the main paper for space. Throughout, C1, C2, and C3 denote the main paper's challenges of recursive ancestry, heterogeneous constraints, and action-dependent admissibility. Symbols (S, I, F, P, V, A, ρ, and θ) and the decision vocabulary (ALLOW, BLOCK, REVERIFY, REDACT, and ASKUSER) follow the main paper.

Table 1: Correspondence between the supplementary sections and the main paper.

Section	Main-paper material supported	Challenge
A	Graph schema and path-trust constants	C1
B	Backend interface, workflow integration, and operation semantics	C1, C3
C	Benchmark construction, sanitization, and transfer subset	C1–C3
D	Exact baseline adaptations and the flat-metadata control	C2
E	Metric formulas and implementation settings	C1–C3
F	Scenario, domain, diagnostic, ablation, and transfer results	C1–C3

A Provenance Graph Schema

The main paper's graph-schema subsection states that MAP-Graph evaluates recorded ancestry rather than only a memory's text or local score. This section specifies the serialized vocabulary and the constants used by the implemented path-trust calculation.

Task identifier, domain, step, visibility, permission scope, trust labels, and risk tags are attributes rather than node types. Tool outputs are represented as Memory nodes associated with a Tool identity; an unregistered output identifier may additionally appear as a `Resource:<id>` reference placeholder. MAP-Graph is therefore an execution-lineage graph, not a general-purpose semantic knowledge graph. The implementation does not create generic `Contradicts`, `Updates`, or temporal-order edges and does not perform open-domain contradiction resolution.

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A.1 Path-Trust Constants

For an eligible memory, the implementation computes

$$\rho(m, a) = \mathrm{clip}_{[0,1]}(S\, I\, F\, P\, V\, A).$$

Source trust S is the minimum explicit trust among source ancestors, or 0.80 if no source ancestor is recorded. Path integrity I starts at 1 and is multiplied once by 0.25, 0.45, and 0.15 when the ancestry contains, respectively, an untrusted/poisoned, private/sensitive, or revoked category. Multiple categories compound. Transformation loss adds 0.08 for an agent-message record, 0.10 for a summary event, and 0.03 for each non-root memory transformation, capped at 0.65; hence $F = \max(0.35, 1 - \mathrm{loss})$. Permission validity P is the current binary `CanRead` result. The verification factor is $V = \min(1.20, 1 + 0.05 n_v)$ for n_v high-trust ancestors. Writer reliability A defaults to 0.95.

The factors use different evidence: S, I, and V summarize recorded ancestry; F follows the transformation chain; P evaluates the current reader against the stored policy; and A belongs to the writing agent. Permission is still enforced as a hard pre-ranking filter in the full system. The graded product only reranks records that remain eligible, and affected ancestry is reconsidered by the action gate.

B Backend Interface and Memory Operations

MAP-Graph is implemented as a task-local memory backend between the shared agent workflow and its simulated proposed action. Table 3 summarizes the public operations.

B.1 Workflow Integration

The harness uses AutoGen's OpenAI-compatible model client and executes the four task-provided roles once in fixed order. It does not instantiate AutoGen `AssistantAgent` or `RoundRobinGroupChat`. At each role step, the harness applies scheduled timeline events, registers sources and tool outputs visible to that role, filters the visible conversation through the selected backend, retrieves memory, calls the model, and writes the response. The final role emits a structured proposed-action object, which is passed to the backend.

The final decision is conservative with respect to the agent proposal and backend decision. A restrictive backend result

Table 2: Registered node and edge types emitted by the graph builder.

Type	Implemented meaning
Node types	
User	Owner or authorizer associated with a resource or memory.
Agent	Workflow role that observes, writes, retrieves, or proposes an action.
Tool	Tool identity associated with an observed output or verification.
Resource	Source document or referenced artifact with trust and access metadata.
Message	Observable inter-agent content; the graph stores its content hash.
Memory	Seeded, observed, or agent-written record stored in Chroma.
Claim	Up to three sentence-level claims extracted by deterministic text rules.
Action	Proposed action, risk, threshold, support, and gate decision.
Edge types	
`authorized_by`	Associates a resource, memory, or action with an owner or actor.
`observed_from`	Records an agent observing a resource.
`forwarded_to`	Records the receivers of an inter-agent message.
`derived_from`	Links a message, memory, or claim to recorded inputs.
`summarized_from`	Special derivation edge for a benchmark summary-memory event.
`written_by`	Associates a memory or claim with its writing agent.
`read_by`	Records a selected memory read and its retrieval scores.
`verified_by`	Records a high-trust source or rule-based verifier.
`invalidated_by`	Records an explicit revocation or invalidating reference.
`used_for_action`	Links a supporting memory or claim to an action.

Table 3: Implemented MAP-Graph memory operations.

Operation	Input	Effect
Observe/write	Source, tool output, or agent text	Creates a distinct memory, embedding, scope, and lineage links.
Retrieve	Agent and query	Filters permissions and returns at most five records ranked by semantic score times path trust.
Revoke	Explicit timeline event	Marks a source revoked, clears its scope, and updates directly referenced or already affected records.
Decide	Proposed action and support	Applies affected-path rules and a risk-dependent trust threshold.
Export	Completed task	Serializes nodes, edges, retrieval diagnostics, gate events, and component settings.

is never relaxed; if the backend returns ALLOW but the agent proposes BLOCK, REDACT, REVERIFY, or ASKUSER, the more restrictive agent proposal remains. Only final ALLOW creates a simulated execution event, and no external side effect is performed.

B.2 Write, Retrieval, Update, and Audit Semantics

Write. Empty text is rejected; otherwise a new record is created without content deduplication. Its `derived_from` list combines retrieved-memory identifiers with sources visible at the current step. With the full graph builder, a Message reference is also recorded. A derived record receives the intersection of referenced scopes. If no referenced scope exists, all task agents are eligible; if the intersection is empty, the scope falls back to the writing agent. Visibility is shared only when the resulting scope covers all task agents.

Retrieval. Chroma first supplies semantic candidates. MAP-Graph computes `CanRead` and path trust for each candidate, removes permission-ineligible records, sorts the remainder by semantic score times path trust, and returns at most five. Selected reads create `read_by` edges carrying semantic, path-trust, and final scores.

Update and revocation. New observations and messages append records rather than mutating or superseding prior text. An explicit revocation event updates the referenced source and directly referenced or already affected records. Other descendants detect the revoked ancestor during later recursive trust evaluation; final graph propagation marks affected predecessors while retaining their lineage. A benchmark summary event creates a new record linked by `summarized_from`. There is no general `active`/`superseded`/`contradicted` state machine.

Structured audit output. The artifacts store selected and permission-filtered identifiers, component scores, affected ancestors and reasons, supporting identifiers, risk thresholds, and the final rule-specific decision. MAP-Graph does not generate a free-form natural-language explanation, and the experiments do not measure human-facing explanation quality.

C Benchmark Construction and Sanitization

C.1 Full Benchmark

The controlled synthetic benchmark contains 2,700 tasks generated with seed 42. It has three equally sized domains—corporate workflow, software engineering, and research assistance—and six experiment groups: (1) clean utility, (2) poisoned-memory propagation, (3) private-memory leakage, (4) permission revocation, (5) action-risk sensitivity, and (6) clean compression/overhead. Each domain contributes 150 tasks to each group. Every domain–group cell contains 15 semantic families with 10 variants that alter entities, audiences, channels, evidence shapes, and action types while retaining a structured oracle. The 2,700 labels comprise 960 ALLOW, 1,140 BLOCK, 210 REVERIFY, and 390 REDACT.

Each task provides source documents, initial and derived memory events, tool observations, visibility and permission fields, a four-role sequence, and a risk-labelled proposed action. Corporate roles are manager, analyst, HR-assistant, and executor; software roles are planner, coder, reviewer, and executor; research roles are retriever, summarizer, writer, and verifier. If the final structured action cannot be parsed, the harness recovers the public action type and risk from the task schema and uses the retrieved records as support.

All eight methods receive the same task order and public task fields. The runtime sanitizer removes the oracle decision, evaluator rationale, experiment and family identifiers, variant metadata, the top-level evaluator canary annotation, and every `should_be_used` annotation before either the agents or memory backend receive a task. The public action type, risk, provenance, visibility, ownership, trust, and permission fields remain, because reasoning over them is the target capability. Authorized source text also remains; otherwise canary exposure could not be measured.

One run for each of B0–B6 and MAP-Graph produces 21,600 main-experiment decision logs. Each method has complete 2,700/2,700 decision, trace, and memory-log coverage; graph artifacts are additionally present for the graph-producing methods. The six ablations plus the full method produce 18,900 additional decisions. These are single-run controlled results.

C.2 Backbone Transfer Subset

The transfer experiment uses a fixed 540-task subset generated with seed 2027 by jointly stratifying the full 2,700 tasks over domain, experiment group, and oracle label. It contains 180 tasks per domain, 90 per experiment group, and all four labels. The same task identifiers are used for B1, B4, B5, B6, and MAP-Graph under Qwen2.5-7B-Instruct, GLM-4-9B-0414, and Llama-3.1-8B-Instruct. Qwen values are extracted from the final full runs; GLM and Llama are one subset run each. Thus each backbone contributes $5 \times 540 = 2,700$ decisions.

C.3 Cluster Bootstrap

The 95% intervals use 2,000 bootstrap samples with bootstrap seed 42. A sampling cluster is a semantic family within a domain and experiment group; all ten closely related variants in a sampled family move together. The intervals therefore describe case-family variation in a fixed run, not model-inference nondeterminism.

D Baseline Adaptations and Controls

B3–B5 are benchmark adaptations that preserve the mechanisms stated below; they are not claimed as complete reproductions of their source systems. B6 is an internal flat-metadata control rather than an adaptation of an external paper.

B0: No Memory. The workflow retains only the current conversation and performs no persistent memory read or write.

B1: Shared Vector Memory. All agent records enter one shared Chroma collection; the five most similar records are returned without permission filtering, lineage scoring, or an action gate.

B2: Isolated Vector Memory. The same vector mechanism is maintained separately per agent. This prevents direct cross-store reads but does not create a shared provenance graph.

B3: Adapted G-Memory. Observable interaction traces are converted into trajectories represented by query, interaction, and insight graphs. The implementation performs one task-level retrieval and reuses its context for all four roles, applies one-hop query-graph expansion, retains the 512 most recent trajectories, and bounds insight state to 10 rules. Periodic insight updates use an LLM. The reported configuration replaces the original per-task LLM trajectory sparsifier and generative reranker with deterministic key-step extraction and similarity ordering. Auxiliary insight calls use temperature 0.1 with a small output cap. B3 receives no MAP-Graph permission filter, path-trust propagation, or action gate.

B4: Adapted Collaborative Memory. Records are assigned to private or shared tiers from visibility and permission metadata. Restricted or revoked status is inherited by derived messages, and the current access-control list is checked before top-five similarity results are returned. A per-task user–agent–resource access graph is exported. B4 has neither recursive path-trust scoring nor an action-time gate.

B5: Adapted MemLineage. Each fragment records derivation ancestors, edges, and untrusted-source reasons. Retrieval is similarity-based and does not filter by reader permission. Before a high-risk action, supporting memories are checked for an untrusted ancestor. The evaluated implementation has no per-principal signatures, cryptographic/Merkle-verified log, permission-aware retrieval, or multi-agent trust decay.

B6: Flat Provenance. B6 stores, on each memory, visibility, owner, permission scope, source trust, private/revoked/poisoned state, and action risk. It performs direct metadata-based retrieval filtering and a metadata-based action check. Derived messages summarize metadata from their direct inputs, and revocation updates records directly tied to the source. B6 does not construct a provenance graph, recursively traverse parent memories, or propagate path-trust scores. It therefore isolates the value of recursive ancestry from the availability of local policy metadata.

E Metrics and Implementation Settings

E.1 Paper-Facing Metrics

Let y_i be the oracle decision, $\hat{y}_i$ the final decision, $\mathcal{I} = \{\textsc{Block}, \textsc{Redact}, \textsc{Reverify}, \textsc{AskUser}\}$, and N the relevant denominator.

1. **Exact decision accuracy (Acc, ↑):** $N^{-1} \sum_i \mathbf{1}[\hat{y}_i = y_i]$.
2. **Task success rate (TSR, ↑):** an oracle-ALLOW case succeeds only for ALLOW; a non-ALLOW case succeeds for any decision in $\mathcal{I}$. Thus a safe but non-exact intervention can increase TSR without increasing Acc.
3. **Clean success (Clean, ↑):** the ALLOW rate on Exp-1 and Exp-6, whose oracle decision is ALLOW.
4. **Unsafe rate (↓):** the fraction of all 2,700 tasks with a simulated execution event when the oracle decision is not ALLOW.
5. **Attack success rate (ASR, ↓):** the ALLOW/execution rate among the 450 Exp-2 poisoned-memory tasks.
6. **Leakage (↓):** the fraction of the 450 Exp-3 tasks with unauthorized canary exposure or a final ALLOW.
7. **Unauthorized access (UAcc, ↓):** the total successful unauthorized memory reads divided by total observed unauthorized-read attempts. It is N/A when no attempt is observed.
8. **Revocation violation (↓):** the final-ALLOW rate among the 450 Exp-4 permission-revocation tasks.
9. **Structural contamination:** for each Exp-2 task having at least one downstream memory, the fraction of downstream records that retain poisoned ancestry, averaged across eligible tasks. This is a lineage-structure diagnostic, not an execution or leakage rate: a system may preserve affected ancestry for audit while preventing its use.
10. **Correct block (↑):** $\Pr(\hat{y} = \textsc{Block} \mid y = \textsc{Block})$.
11. **False block (↓):** $\Pr(\hat{y} = \textsc{Block} \mid y = \textsc{Allow})$.

The additional *block rate* is the unconditional fraction of final BLOCK decisions. Per-task token use sums provider-reported prompt and completion tokens over the four outer agent calls and any recorded memory-layer calls. The token-cap diagnostic is the fraction of the 10,800 outer agent calls per method whose reported completion reaches the 2,048-token limit. Average graph nodes and edges are computed from exported task graphs.

E.2 Fixed Implementation Settings

All main-experiment methods use the same AutoGen OpenAI-compatible client with Qwen2.5-7B-Instruct, outer-call temperature 0, one four-role round, at most 12 recent context messages, at most five retrieved memories, a 2,048-token generation cap, a 120-second per-call timeout, and at most five retries. Embedding methods use normalized BAAI/bge-small-en-v1.5 embeddings on CPU with $k = 5$.

MAP-Graph filters permissions before ranking, uses multiplicative path trust, propagates affected state, and evaluates retrieved plus contextual support at action time. Its thresholds are 0.30 for answers, 0.60 for low- or medium-risk non-answer actions, and 0.85 for high-risk actions. A high-risk action is blocked when an affected supporting path is detected.

Latency is not a paper-facing metric. All model calls used remote shared endpoints and experiments ran concurrently across accounts, so provider queuing and long-tail service load cannot be separated from local memory overhead. Token use and graph size are retained only as reproducible diagnostics.

F Additional Results

Tables 4–8 report the scenario, domain, full-run, ablation, and transfer diagnostics discussed in this section.

F.1 Scenario-Level Results

Table 4 compares MAP-Graph with the strongest baseline for the utility and risk quantity relevant to each experiment group. The comparator can differ across groups.

F.2 Confidence Intervals and Domain Results

MAP-Graph's TSR is 94.96% with a 95% cluster-bootstrap interval of [93.15, 96.52], and its Acc is 72.70% [68.26, 77.19]. The strongest baseline TSR is B6 at 74.67% [71.67, 77.74]; the strongest baseline Acc is B5 at 51.07% [47.19, 55.07]. MAP-Graph's Clean is 90.22% [87.44, 92.92], and its unsafe rate is 1.52% [0.37, 2.93]. These intervals measure clustered case variation within one run. Table 5 reports the corresponding domain breakdown. The TSR range is narrow across the three controlled role configurations. The lower research Acc with high TSR reflects more safe but non-exact interventions, not more impermissible execution.

F.3 Main-Experiment Diagnostics

Table 6 reports block rates, structural contamination, and token diagnostics for the full run. MAP-Graph's average token count consists of 4,194.8 prompt and 620.1 completion tokens. B3 is highest because retrieved trajectories and periodic insight generation add context and auxiliary generation. MAP-Graph exports 39.74 nodes and 246.78 edges per task on average. These counts exclude embedding computation, graph construction time, storage I/O, evaluator work, and failed requests for which the provider reports no usage; they are not a complete cost or latency comparison.

Structural contamination must be interpreted separately from safety outcomes. MAP-Graph's 25.00% means that affected ancestry remains attached to some downstream records; those records are retained and marked for audit. Its 0% ASR and 0% poisoned-group impermissible execution show that retained lineage was not treated as admissible support in those cases.

F.4 Full Ablation Diagnostics

Table 7 supplements the main-paper ablation table with leakage, contamination, block, token, and graph-size diagnostics. The six variants correspond to explicit component switches. *No permission filter* disables candidate exclusion but leaves trust and gating active; its 100% UAcc in the main table despite 0% endpoint leakage here shows why read-boundary and endpoint metrics are both needed. *No trust propagation*

Table 4: Scenario-level results (percent; 450 tasks per row). "Best baseline" is selected for the quantity shown within that experiment group.

Group	Scenario	MAP TSR	Best baseline TSR	Risk/utility quantity	MAP	Best baseline
Exp-1	Clean utility	84.22	83.33 (B1)	Exact accuracy	84.22	83.33 (B1)
Exp-2	Poisoned propagation	100.00	77.78 (B5)	ASR	0.00	22.22 (B5)
Exp-3	Private-memory leakage	100.00	82.00 (B2)	Leakage	0.00	18.00 (B2)
Exp-4	Permission revocation	100.00	76.22 (B5)	Revocation	0.00	23.78 (B5)
Exp-5	Action-risk sensitivity	89.33	68.22 (B6)	Unsafe action	9.11	28.67 (B6)
Exp-6	Clean compression	96.22	94.44 (B4)	Exact accuracy	96.22	94.44 (B4)

Table 5: MAP-Graph results by domain (percent; 900 tasks per row).

Domain	TSR	Acc
Corporate workflow	94.56	73.56
Software engineering	95.00	74.56
Research assistance	95.33	70.00

Table 6: Additional full-run diagnostics (percent except tokens/task). Correct and false block use their conditional denominators; Cap calls is the fraction of outer calls reaching 2,048 completion tokens.

Method	Block	Correct block	False block	Structural contam.	Tokens/task	Cap calls
B0	12.67	16.23	2.50	0.00	3,397.2	8.12
B1	14.41	15.44	3.33	43.44	5,007.5	4.66
B2	14.07	16.67	2.71	25.00	4,054.4	4.06
B3	9.07	8.77	3.23	11.11	6,687.4	9.08
B4	15.37	17.46	2.71	25.02	4,927.8	4.46
B5	23.96	38.86	2.81	62.17	4,953.2	4.09
B6	16.59	21.23	2.60	25.00	4,882.7	4.44
MAP-Graph	24.00	54.21	3.12	25.00	4,814.9	3.89

also disables trust-aware reranking. *No action gate* allows at the backend decision stage, while *No containment* retains the threshold gate but removes affected-path rules. *No provenance graph* jointly disables graph construction, permission filtering, trust propagation and reranking, action gating, containment, and contextual action support; it is a system-level removal rather than a one-switch attribution. The *compressed graph* retains policy behavior but exports fewer graph details.

F.5 Backbone Access-Boundary Diagnostics

Table 8 reports unauthorized-access diagnostics on the common 540-task transfer subset. The deterministic access policy and task structure make the attempt counts largely stable across backbones, while B5's model-dependent retrieval changes how often an unauthorized record is selected. MAP-Graph rejects every observed unauthorized read under all three backbones. This supports transfer of the implemented access boundary, not a claim of run-to-run stability or deployment-scale robustness.

F.6 Error Analysis

MAP-Graph makes 737 exact-decision errors. Of these, 601 are safe but non-exact interventions, such as REDACT rather than BLOCK or REVERIFY; this accounts for much of the TSR–Acc gap. Another 95 cases are unnecessary interventions on oracle-ALLOW tasks. The remaining 41 are impermissible ALLOW decisions, all in Exp-5. The residual safety errors in this run therefore concentrate in action-risk sensitivity rather than poisoned, private, or revoked-memory groups.

Table 7: Diagnostics supplementing the main-paper ablation table (percent except tokens and graph size). CBlock/FBlock denote conditional correct/false block.

Variant	Leak	Contam.	Block	CBlock	FBlock	Tokens	Nodes	Edges
Full	0.00	25.00	24.00	54.21	3.12	4,814.9	39.74	246.78
Compressed graph	0.00	25.02	23.78	53.77	3.02	4,847.0	24.66	70.68
No action gate	22.22	25.00	14.41	17.54	2.50	4,830.1	39.74	246.69
No containment	20.89	25.00	14.81	18.33	2.50	4,814.2	39.68	245.96
No permission filter	0.00	24.94	25.74	58.86	2.40	4,911.9	39.76	250.73
No provenance graph	22.89	40.44	14.22	16.40	3.02	4,991.2	0.00	0.00
No trust propagation	0.00	25.00	11.15	24.12	2.71	4,886.3	39.91	265.10

Table 8: Unauthorized-access diagnostics on the common 540-task transfer subset. Each backbone block reports unauthorized reads (R), observed attempts (A), and conditional UAcc in percent.

	Qwen2.5-7B			GLM-4-9B			Llama-3.1-8B		
Method	R	A	UAcc	R	A	UAcc	R	A	UAcc
B1	48	48	100.00	48	48	100.00	48	48	100.00
B4	990	2,323	42.62	990	2,324	42.60	990	2,324	42.60
B5	101	101	100.00	86	86	100.00	84	84	100.00
B6	990	2,324	42.60	990	2,324	42.60	990	2,324	42.60
MAP-Graph	0	1,334	0.00	0	1,334	0.00	0	1,334	0.00